Aurintricarboxylic acid as a basis for inhibiting Cdc14 phosphatase in *Alternaria Alternata*

Wang J, Yu R, Zhu C

Abstract

Cdc14 (cell division control 14) phosphatase is crucial in the pathogenesis and morphogenesis of certain fungi (17). Cdc14 is highly conserved within the fungal kingdom, hence, it poses a potential target in *Alternaria Alternata*, a fungal pathogen responsible for black rot and implicated in agricultural yield losses (5, 18). Aurintricarboxylic acid is capable of inhibiting Cdc14 from *A. Alternata*. Moreover, aurintricarboxylic acid is effective against other fungal species such as *Penicillium digitatum*, *Ustilago hordei*, and *Cercospora zeina*, acting as a potential chemical for use in general fungicides. Alterations to the structure of aurintricarboxylic acid can increase its affinity for Cdc14 phosphatase, increasing the efficacy of aurintricarboxylic acid as an inhibitor.

Introduction

World hunger is a problem that is faced by over 1 in 9 people across the world (1). In its current state, society cannot support the existing population, much less the ever-increasing future population (2). Fungal pathogens threaten agriculture around the world, contributing to reduced agricultural yields and crop shortages (3). One such fungal pathogen is *Alternaria alternata*. *Alternaria* is a fungus under the *Ascomycota* phylum and causes more than 20% of agricultural losses (4). *Alternaria* species are known to colonize numerous crops including cereals, tomatoes, cucumbers, and citruses, causing black rot or citrus brown rot and rendering crops unsafe for human consumption (5).

Current fungicides for *A. alternata* include azoxystrobin, trifloxystrobin, acibenzolar, iprodione, chlorothalonil, captan, copper hydroxide, pyraclostrobin, mancozeb, and pyrimethanil (6). However, all of the stated fungicides are irritants and environmental hazards, with several being toxic to humans (7-16). The purpose of this study is to discover a potential fungicide that is not detrimental to human health, hence mitigating agricultural damage with no unintended consequences.

This research paper explores the development of an inhibitor for Cdc14 phosphatase (Cdc14p), a dual-specificity phosphatase (DSP) crucial for pathogenesis and morphogenesis observed in *Fusarium graminearum* (17). Cdc14p is highly conserved within the fungal kingdom, and is, thus, implicated in *Alternaria Alternata* pathogenesis and cell division (18). Therefore, Cdc14p poses a potential target to hinder the growth of *A. alternata*.

One of the biggest challenges of reducing fungal infection is the pathogens' rapid development of resistance to existing fungicides. However, according to Clustal Omega, Cdc14p is highly conserved among the fungal kingdom, meaning the sequence of Cdc14p has not altered significantly through evolution (Fig. 1). Thus, the proposed inhibitor, which targets the conserved domain, might be less impacted by the development of fungicide resistance. Moreover, the Cdc14p orthologs in humans are not essential for cell viability or mitotic exit, hence, utilizing an inhibitor of Cdc14p would have a minimal negative effect on the health of humans

(19). Cdc14p is not present in plants (18). Therefore, through the inhibition of Cdc14p, a fungicide can be made that will counteract Alternaria Alternata's effects on crop yields with minimal negative side effects on crops and humans.

To create an inhibitor for Cdc14p, a plasmid containing the AaCDC14 gene was transformed into chemically competent *E. coli* cells, expressed, and purified. Then, a Bradford Assay, activity assay, steady state assay, and substrate specificity assay were conducted to determine protein concentration, enzyme specific activity, K_M and V_{max} of the enzyme for pNPP, and enzyme specificity for different phosphopeptides, respectively. Based on the results of these experiments, candidate inhibitors generated from high throughput screening were tested, and the best inhibitor was optimized in MOE. Therefore, an effective general inhibitor for Cdc14p in fungi was found, which can be further modified into a better inhibitor and used to create new fungicides to combat *A. Alternata* and other species.

Results

Protein Purification

An SDS-PAGE was performed for solutions collected from the nickel buffer A, 40mM imidazole wash, 100mM imidazole wash, and 250mM imidazole wash effluents in the nickel column chromatography (Fig. 2).

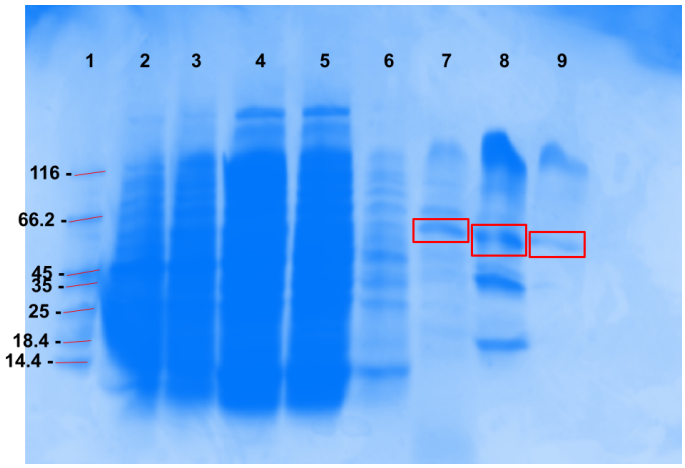

Figure 2 **SDS-PAGE of nickel column chromatography effluents**. Lane 1 is MW, lane 2 is arabinose-induced cells, lane 3 is uninduced cells, lane 4 is cell lysate, lane 5 is nickel buffer A elution fraction, lane 6 is flow through 2, lane 7 is the fraction elected with 40 mM imidazole, lane 8 is the fraction elected with 100 mM imidazole, and lane 9 is the fraction elected with 250 mM imidazole. The bands outlined in red represent AaCdc14. The 100mM elected fraction was used for later experiments.

The MW of AaCdc14 is 52.62 kDa, estimated from the primary sequence. Detectable amounts of desired protein can be found in the 40mM imidazole effluent, 100mM imidazole effluent, and 250mM imidazole effluent, with the most significant amount residing in the 100mM imidazole effluent. Smearing is seen in lanes 7, 8, and 9, likely due to protein overloading. Protein degradation is also detected, with 2 dark bands of lower MW being present in lane 8.

The 100mM imidazole effluent and 250mM imidazole effluent were run through SDS-PAGE in a second trial at lower concentrations (Fig. 3) to prevent smearing. The resulting gel had a clear band of the desired protein between the 45 kDa and 66.2 kDa markers.

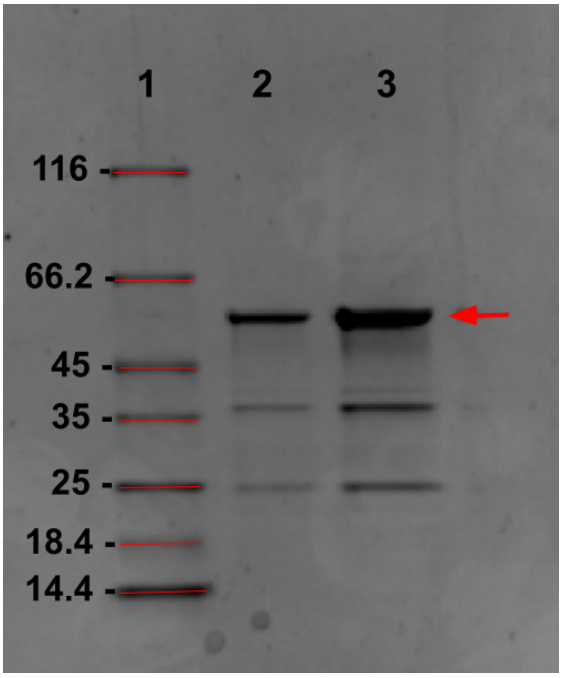

Figure 3 **SDS-PAGE of 100mM imidazole effluent and 250 mM imidazole effluent**. Lane 1 is the molecular weight, lane 2

is the fraction elected with 250 mM imidazole, and lane 3 is the fraction elected with 100 mM imidazole. The red arrow points to the band for AaCdc14.

The protein concentration of the fraction elected with 100 mM imidazole was determined. Three trials of a 20x protein dilution Bradford Assay yielded an average absorbance of 0.38 ± 0.05, which was converted to a protein concentration of 114.63 ± 46.74 µM using a BSA standard curve.

The nanodrop spectrometry of the 20x protein dilution yielded an absorbance of 0.505. Using the theoretical extinction coefficient of 69790, approximated from the AaCdc14p sequence, a protein concentration of 144.72 µM was determined.

The average of the results from the two methods was 129.68 µM, which was determined to be the purified protein concentration.

Protein Characterization

In order to visualize AaCdc14, a homology model was built based on the template protein ScCdc14 (PDB ID: 5XW5) (20) in MOE (Fig. 4). The figure shows that the homology model has a high level of agreement with the template model, as indicated by the green areas. Dissimilar residues were deleted from the N-terminus and C-terminus of the query sequence.

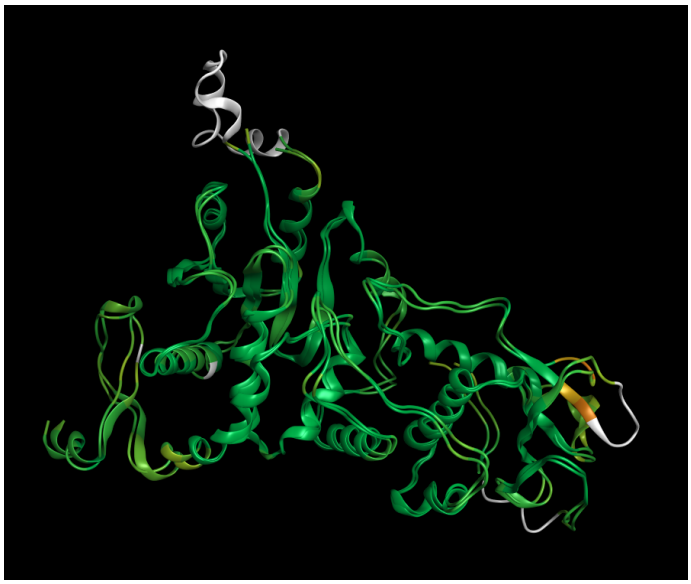

Figure 4 **Overlay of AaCdc14p homology model with the experimentally determined structure of ScCdc14, colored by RMSD**. Green indicates similar, red indicates dissimilar, and white indicates no overlap. The average RMSD is 1.16.

In the generated ligand interaction map of the homology model, Cys330 interacts with the phosphate group. In comparison, in the ligand interaction map of the ScCdc14, Ser283 interacts with the phosphate group (Fig. 5). The wild-type ScCdc14 has a cysteine rather than a serine residue, but the original cysteine must be changed to a serine to capture the crystal structure of ScCdc14 as cysteine reacts too quickly to be captured in a crystal structure (20).

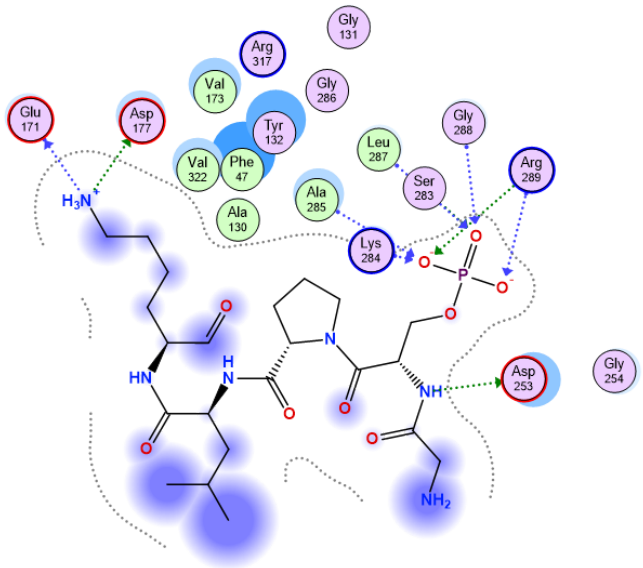

Figure 5.1 **Ligand interaction map of the tetrapeptide (pSPLK) in the active site of ScCdc14**. Serine283 was mutated from a cysteine found in wt-ScCdc14.

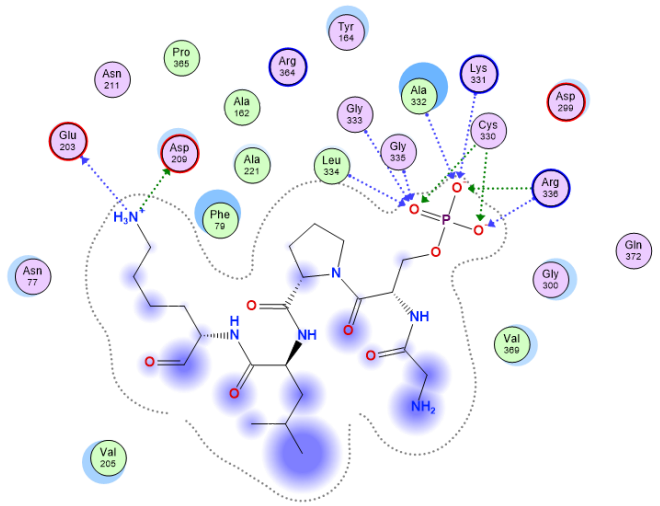

Figure 5.2 **Ligand interaction map of the tetrapeptide (pSPLK) in the active site of AaCdc14**.

There is likely an arginine 6 residues downstream of the cysteine that also interacts with the phosphate in the active site in AaCdc14 similar to ScCdc14. The conservation of the arginine and cysteine in both proteins insinuates the importance of Cys330 and Arg336 in AaCdc14 with ligand interactions.

pNPP Specific Activity

To determine if the phosphatase activity from the purified protein was due to AaCdc14 and not contaminating PPPs from the *E. coli* used to express AaCdc14, a tungstate inhibitor, which inhibits PTPs, was tested against AaCdc14 (21). The average concentration of pNP in the assay wells without inhibitor was 31.91194969 µM while the average concentration of pNP in the assay wells with sodium tungstate inhibitor was -2.018867925 µM (extrapolated to be 0 µM). The lack of activity confirms AaCdc14 to be the main source of phosphatase activity in the purified protein samples.

The continuous assay results indicated a linear activity rate over the full 60 minutes for enzyme concentrations of 3 µM and lower (Fig. 6).

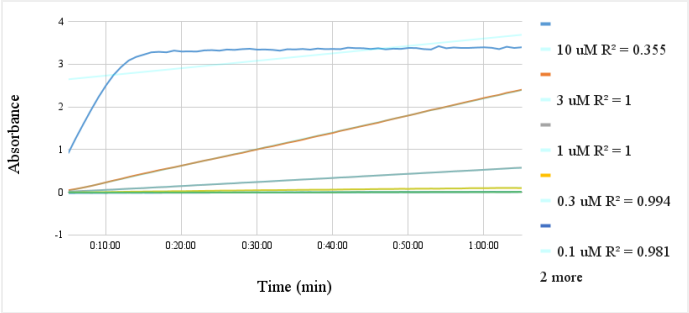

Figure 6 **AaCdc14 linearity assay**. Different concentrations of enzyme (10 µM, 3 µM, 1 µM, 0.3 µM, 0.1 µM) were run in a continuous assay for 60 minutes and the absorbance was measured. The results showed a linear rate of change in absorbance for 3 µM and lower concentrations.

Substrate Specificity

24 different phosphopeptides were tested to determine the best substrate. As shown in Fig. 7, the most effective phosphopeptide with the highest K_{cat}/K_M was +2K+4R (#22), +4K (#15), and +4R(#16). These peptides had a phosphorylated serine, a proline at the +1 position, and a lysine at the +3 position. #16 and #22 shared an arginine at the +4 position. The results matched the prediction from the ligand interaction map (Fig. 5.2) and were consistent with the results in a previous paper that used Cdc14 in other fungal species (18).

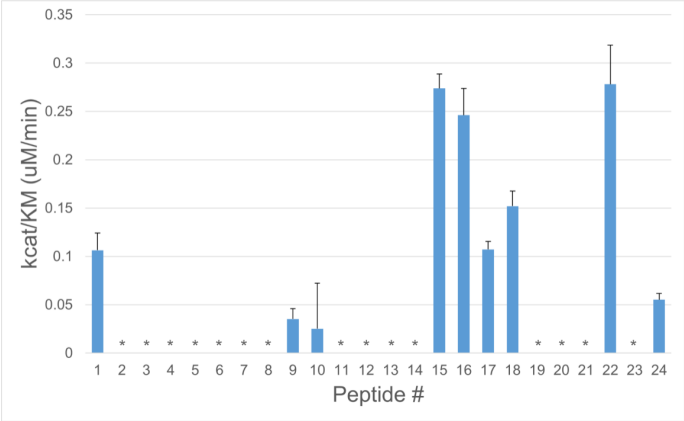

Figure 7 **k_{cat}/K_M (µM/min) for AaCdc14 towards different phosphopeptide substrates**. A Biomol green assay was performed to determine the k_{cat}/K_M of the dephosphorylation of different phosphopeptides to determine the best substrate for AaCdc14. The error bars record the standard deviation of the three trials and the asterisk marks peptides with negligible activity. Enzyme concentration was 0.2µM, and the experiment

ran 10 minutes at room temperature. The sequences of the peptides are shown in Table 1.

Peptide #	Sequence	Peptide #	Sequence
1	HT{pS}PIKSIG	13	HT{pS}PIRSIG
2	HT{pT}PIKSIG	14	HT{pS}PILSIG
3	HT{pY}PIKSIG	15	HT{pS}PIKKIG
4	HT{pS}AIKSIG	16	HT{pS}PIKRIG
5	HT{pS}GIKSIG	17	HT{pS}PIKSKG
6	HT{pS}GIASIG	18	HT{pS}PIKSRG
7	HT{pS}GIKRIG	19	HT{pS}PKASIG
8	HT{pS}PKKSIG	20	HT{pS}PIAKIG
9	HT{pS}PRKSIG	21	HT{pS}PIASKG
10	HT{pS}PEKSIG	22	HT{pS}PKKRIG
11	HT{pS}PPKSIG	23	RT{pS}PIKSIG
12	HT{pS}PIASIG	24	HD{pS}PIKSIG

Table 1 **Peptide sequences from the substrate specificity assay**.

Steady-State Kinetic Parameters

To determine the K_M and V_{max} of AaCdc14 for pNPP, varying substrate concentrations were tested to create a Michaelis-Menten curve. The K_M and V_{max} of the studied enzyme are 41.3±0.00706 mM and 0.00858±0.00573 mM/min, respectively, as determined by the Michaelis-Menten graph (Fig. 8).

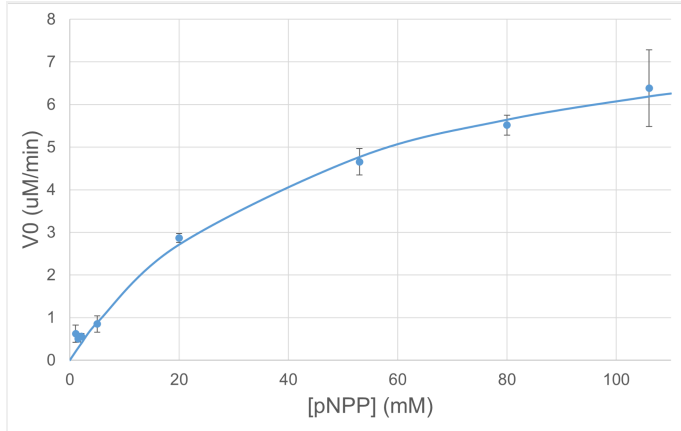

Figure 8 **Enzyme activity assay**. Initial velocity vs. substrate concentration was used to determine V_{max} and K_M of the enzyme by minimizing squared error in the Michaelis-Menten equation with V_{max} and K_M as variables. V_{max} was found to be 0.00858± 0.00573 mM/min and K_M was found to be 41.3±0.00706 mM (R^2 = 0.9923).

The assay was necessary for determining a good pNPP concentration to use for the inhibitor assay.

Inhibitor Assay

Using the pNPP standard curve along with the calculated K_M of AaCdc14 for pNPP, a test was set up to determine which inhibitor in a set of 11 pre-screened substrates (Fig. 9) would be most effective in inhibiting AaCdc14. Having a substrate concentration below the K_M of 41.3 determined from the steady state assay ensured that the substrate did not outcompete the inhibitors. Of the inhibitors, I1 and A9 had the best inhibition, with H7, G5, and G6 being moderately effective. H7 had similar inhibition to I1 and A9, but it was tested at 1 mM rather than 100 µM. The rest of the inhibitors, E1, I2, D3, D7, and I9 (100 µM), were not effective.

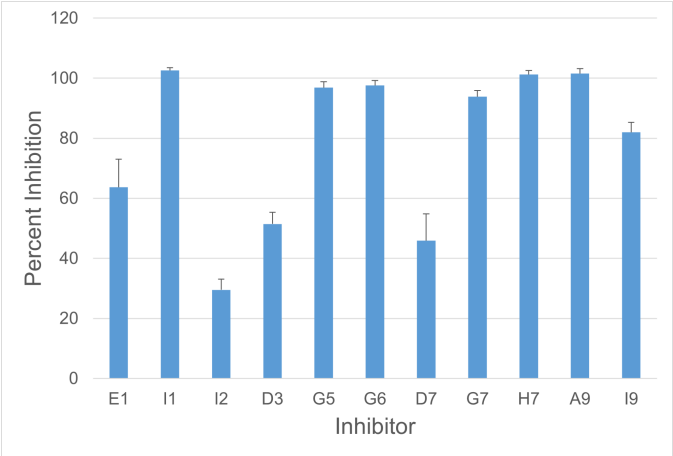

Figure 9 **Inhibitor percent inhibition assay**. The activity of AaCdc14 inhibited by different inhibitors was measured via pNP concentration. The Z score of the assay was 0.6172023096. The error bars represent the standard deviation of the three trials.

Inhibitor Docking in MOE

The 11 inhibitors were docked into the active site of the AaCdc14 homology model in MOE to visualize interactions *in silico*. Interactions to the conserved Cys330 and Arg336 residues were specifically noted (Table 2).

Inhibitor	**Best S score**	**Hbond to Cys330 & Arg336**
I1	-6.2660	Yes
D7	-6.2288	No
E1	-6.1952	No
A9	-5.8700	Cys330 only
G5	-5.7133	Yes
G7	-5.4955	No
D3	-5.2448	No
I9	-4.9083	Yes
H7	-4.8788	No
I2	Did not dock	N/A
G6	Did not dock	N/A

Table 2 **Inhibitor S scores and active site interactions**. 20 poses for each inhibitor were generated in MOE and the top 5 poses were compared. The table shows the S score of the best pose and whether the inhibitor hydrogen bonds to Cys330 or Arg336.

Comparing the results from the inhibitor assay *in vitro* and the inhibitor docking *in silico*, inhibitor I1 was selected as the ideal inhibitor for AaCdc14 as it had the highest inhibition percentage in the inhibitor assay, the highest S score in MOE, and hydrogen bonded to both Cys330 and Arg336 according to the best docking pose in MOE.

Determining IC_{50} of I1

Absorbances collected from 10 different inhibitor concentrations of I1 were fitted to a four-parameter logistic equation (Fig. 10). The resulting IC_{50} for inhibitor I1 was calculated to be 1.95354 µM.

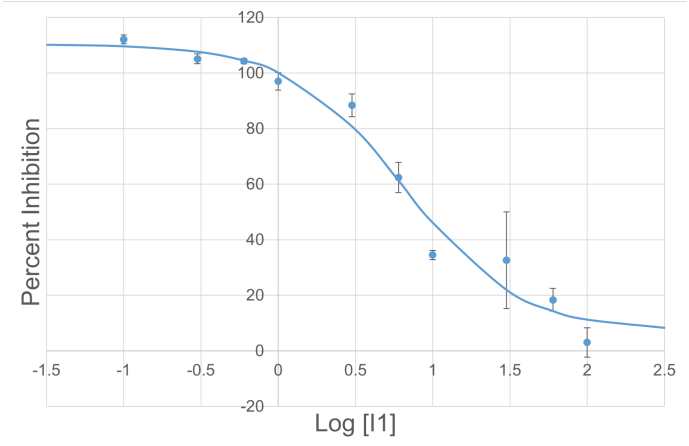

Figure 10 **Inhibitor I1 four-parameter logistic curve for IC_{50} determination**. The average of three trials was fitted into the four-parameter logistic equation (Fig. 16), with the best fit curve (R^2 = 0.9907) providing the following values: maximum absorbance of 0.436894, minimum absorbance of -0.0556981, Hill coefficient of 0.968353, and an IC_{50} of 1.95354 µM. All values were converted to percent inhibition and graphed. The error bars represent the standard deviation between the three trials.

Optimization of I1

The structure of inhibitor I1 was altered in MOE to change the affinity from -5.99 kcal/mol to -7.92 kcal/mol. The Lipinski rule of 5 was followed during modification.

The first change that was made was adding a propyl group that extended into a nearby hydrophobic pocket, as well as mirroring the hydrophobic group in the ScCdc14 crystal structure substrate (-6.53 kcal/mol). The I1 inhibitor already started out with 5 hydrogen bond donors, so if a hydrogen bond donor would work well in another place, a hydrogen bond donor would have to be deleted according to Lipinski's rule of 5.

The second change that was made was turning an alcohol group to an ethyl ether group because the alcohol was originally in a somewhat hydrophobic region and it also freed up a hydrogen bond donor to be put into another place (-6.89 kcal/mol).

The third change that was made was turning a hydrogen into an ethyl alcohol group, allowing the inhibitor to form strong hydrogen bonding interactions with a hydrogen bond acceptor region (-7.41kcal/mol).

The fourth change that was made was turning a carbonyl into an ether because it allowed for easier polar interactions with the oxygen, reducing the log(P) score (-7.61 kcal/mol).

The fifth change that was made was turning a methyl group into a methyl trifluoride, which further increased the hydrophobicity of that group in the hydrophobic pocket and further pronounce the similarities between the ScCdc14 substrate and the competitive inhibitor (-7.80 kcal/mol).

The last change that was made was turning a hydrogen into a methyl group to yet again increase the hydrophobicity of the inhibitor in the hydrophobic pocket (-7.92 kcal/mol, logP of 4.53) (Fig. 11).

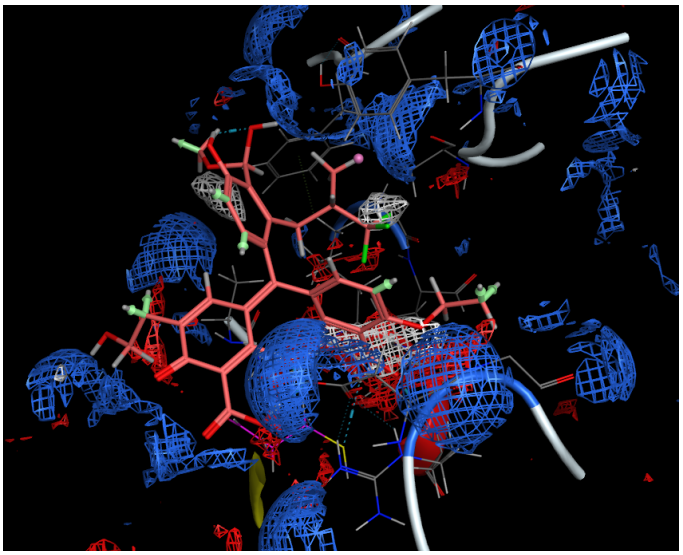

Figure 11.1 **Modified inhibitor docked into AaCdc14 homology model**. The display shows the electrostatic surface map (white is hydrophobic, red is h bond acceptor, blue is h bond donor). The final docking score is -7.92 kcal/mol and the final log(p) is 4.53.

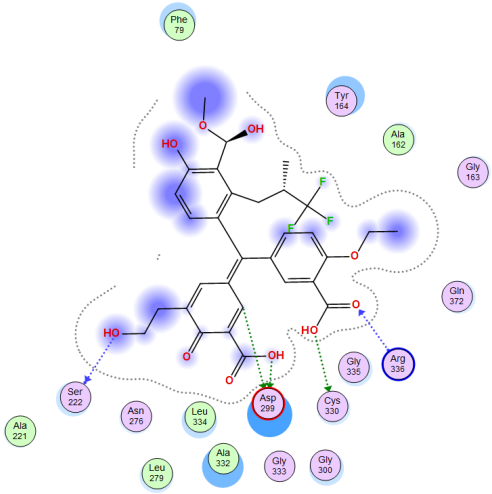

Figure 11.2 **Modified inhibitor interaction map with AaCdc14 pocket**.

Discussion

The substrate specificity result of AaCdc14 showed a similar trend with Cdc14 homologs (18). It means that Cdc 14 is highly conserved and the inhibitor we designed can not only inhibit AaCdc14 but broadly inhibit Cdc 14 enzymes.

The substrate specificity assay revealed that the best substrates for AaCdc14 all interacted with cysteine and arginine residues conserved in fungal Cdc14.

Therefore, it is likely that effective competitive inhibitors for AaCdc14 would also interact with the cysteine and arginine residues. Other properties of an effective inhibitor include having a good electrophilic site and the ability to form strong intermolecular bonds that will stabilize docking positions (23).

To fulfill the role of an inhibitor, the inhibitor will most likely covalently bind to and undergo a nucleophilic attack from the conserved cysteine residue. Because the inhibitor may undergo a nucleophilic attack, having a receptive electrophilic site is crucial. However, due to the constraints of MOE, the electrophilicity of the binding site is unable to be modeled accurately. As such, the final inhibitor will not work the same *in vitro* as it does *in silico*. Of the inhibitors tested *in vitro* (Fig. 14), the ones that had a good electrophilic site (generally characterized by having a somewhat stable resonance structure with an atom that can accept a bond) tended to do comparatively better in experimental trials than MOE predicted.

On the other hand, inhibitors that had strong intermolecular interactions tended to do comparatively better in MOE than *in vitro*. Having strong intermolecular interactions at the active site makes docking easier, and thus decreases the S score of the inhibitor, which is the main factor that MOE uses to predict effective inhibitors. Due to feasibility constraints of not being able to synthesize the changed inhibitors, MOE was used to test the final inhibitor.

The best inhibitor, according to both experimental procedures and MOE, was an inhibitor labeled I1 (aurintricarboxylic acid). I1 has many different hydrogen bond donors and acceptors– making it a good inhibitor according to MOE– and also an electrophilic resonance structure that forms both an aromatic ring and tertiary carbocation, although those interactions are not considered in MOE.

Lab results indicated that having a good electrophilic site, a nonpolar side chain, and around 2 rings were optimal for inhibitors of AaCdc14. MOE results suggested that inhibitors with a lot of hydrogen bonds, dipole-dipole forces, and hydrophobic interactions had the best chance of docking.

The I1 inhibitor is already an effective inhibitor, both by the lab and by MOE standards, but the goal was to find a better inhibitor. Another effective inhibitor in the lab, G5 (2-amino-4- (2,3-dichlorophenyl) -7-hydroxy-4H-chromene-3-carbonitrile), had good side chains, electrophilic site, and interacted with conserved cysteine and arginine residues, and it had a significant percent inhibition when tested in the inhibitor assay. G5 had a nonpolar side chain, something that I1 did not have, suggesting that both hydrogen bonds and hydrophobic interactions stabilize the docking of the molecule. As such, to make I1 a better inhibitor, taking advantage of both hydrogen bond interactions and hydrophobic interactions could be effective.

In addition, because the original crystal structure substrate for ScCdc14 had a phosphatase end (polar), and a proline end (nonpolar), having a nonpolar group could prove to be the optimal arrangement for Cdc14 inhibition. Competitive inhibitors are modeled after substrates because they both bind at the active site, and due to the enzyme's evolved fit around the substrate, the substrate should theoretically have a high affinity for the enzyme, which can be taken advantage of to design an effective inhibitor.

One limitation to altering the structure of I1 is changing the ketone functional group as it provides a crucial part of the electrophilic resonance structure that allows the formation of a tertiary carbocation and aromatic ring. While MOE might not be able to predict changes in affinity due to functional groups that play a role in nucleophilic attack, the inhibitor would not be able to function well in an experimental setting.

In attempting to create a better inhibitor than I1 in MOE, it was quickly revealed that MOE does not model feasibility accurately. MOE models how effective an inhibitor is by an affinity score, which is almost only affected by intermolecular interactions. Adding 5 phenyl rings in a row drastically increased the affinity score because of how many hydrophobic interactions it created, but synthesizing the molecule and allowing for an interaction of the electrophilic site on the inhibitor with the nucleophilic cysteine residue in the enzyme would be difficult. All of the changes that were made were justifiable using an intermolecular interaction argument and large functional groups like cycloalkanes and phenyl rings were avoided for the sake of synthesis and not moving the interaction site. The electrophilic site was also not disturbed so as to not destroy its *in vitro* capabilities.

Cdc14 phosphatase is highly conserved in the fungal kingdom, specifically the cysteine and arginine residues in the active site. Because I1 interacts with both the cysteine and arginine residues, it is likely effective in inhibiting other fungal pathogen species. Cdc14 phosphatases in several other fungal species were studied in tandem with AaCdc14, and I1 was found to be highly effective in inhibiting Cdc14p from *Penicillium digitatum*[1], *Ustilago hordei*[2], and *Cercospora zeina*[3]. Inherent enzyme properties like Km, Vmax, and Kcat for the substrate pNPP are also similar between AaCdc14 and ScCdc14, suggesting that active site residues are similar (18). Thus, I1 may be useful for use as a broad-spectrum fungicide to protect crops from fungal diseases.

One drawback of the research used to determine and optimize an inhibitor for *A. Alternata* is that the docking and optimization of I1 were done *in silico* via MOE. While providing a good analysis of the non-covalent interactions and affinity between I1 and the homology model active site, MOE does not take into consideration the electrophilicity of I1 and its potential for covalently bonding with specific residues. Consequently, the docking score in MOE is not a full representation of the inhibitor interactions. In the future, an *in vitro* assay can be performed to assess the effectiveness of the optimized I1 inhibitor. In addition, x-ray crystallography could be performed to generate the true structure of AaCdc14 for a more accurate docking simulation.

Furthermore, aurintricarboxylic acid is an irritant to humans (22). It is unknown whether the molecule modified from aurintricarboxylic acid to have higher efficacy in inhibiting AaCdc14 is an irritant as well. Thus, the designed molecule can be tested to ensure its safety for human health.

Through the transformation and expression of chemically competent *E. coli* with a plasmid containing AaCDC14, a purified AaCdc14 protein was obtained to allow for the testing of effective inhibitors. Using both *in vitro* and *in silico* results, the best characteristics for an inhibitor of AaCdc14 were determined and aurintricarboxylic acid was determined to be the best inhibitor of a set of pre-screened inhibitors from a database of inhibitors. However, because aurintricarboxylic acid is not necessarily an inhibitor specifically made for Cdc14 phosphatase in fungi, alterations were made to amplify the affinity of aurintricarboxylic acid for Cdc14 proteins. The modified structure for inhibiting AaCdc14, as determined in MOE, has a docking energy of -7.92 kcal/mol and retains its electrophilic site.

Experimental Procedure

Transformation and Expression

Frozen *E.coli* BL21 AI cells were thawed on ice and combined with 2 µL of pET-15b plasmids containing the AaCDC14 sequence with a 6xHis tag (Fig. 12). The *E.coli* cells were then incubated on ice for 30 minutes, followed by a heat shock at 42 ˚C for 1 minute. After recovery, 450 µL of 2xYT medium

[1] Jonathan Yu, personal communication, July 17, 2022
[2] Samuel Wilson, personal communication, July 17, 2022
[3] Mia Sosa, personal communication, July 17, 2022

was added to the cells and the cells were incubated at 37 ˚C for 30 minutes.

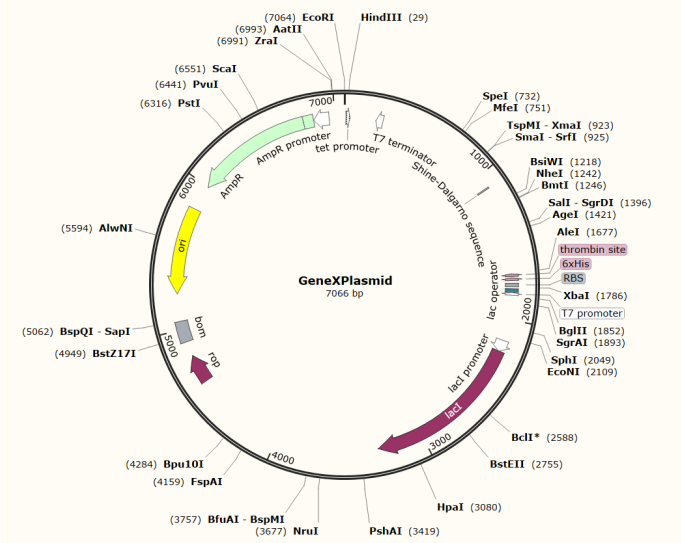

Figure 12 **pET-15b plasmid with AaCdc14 sequence**. The AaCdc14 sequence was inserted between the thrombin site and the T7 terminator in reading frame -1.

The cells were plated on an LB/ampicillin plate and placed in a 37 ˚C incubator overnight. Single colonies were used for inoculation. The cell cultures were mixed with 1 L of 2xYT media and 20% L-arabinose to a final concentration of 0.1% for induction. The solution was incubated in two 2.8-L Fernbach flasks overnight.

Protein Purification

The *E.coli* was put into a lysis buffer (25 mM HEPES pH 7.5, 500 mM NaCl, 0.1% (v/v) Triton x-100, 10 mM imidazole, 10% (v/v) glycerol), 10µM leupeptin, 1µM pepstatin, 4µL universal nuclease, and 1.2mL lysozyme, then incubated on ice for 30 minutes. PMSF was added to a final concentration of 0.5mM to partially inhibit the serine proteases.

The cellular debris was removed by centrifugation at 15,000xg, after which the supernatant was separated for affinity chromatography. 25µL of the supernatant was set aside in a microcentrifuge tube for protein concentration determination and another 15µL was prepared with 2x SDS-gel loading buffer, 1M DTT, heated for 5 minutes at 95˚C, and stored at -20˚C. The affinity chromatography column was prepared with nickel buffer A (25 mM HEPES, pH 7.5, 500 mM NaCl, and 10% (v/v) glycerol) and Ni-NTA beads. The cell lysate was then flowed through the column twice to give the AaCdc14 proteins with the His tags the highest chance of binding.

To isolate the proteins from the column, approximately 30 mL of cell lysate was loaded on the column, and 10 mL of nickel buffer A, 40mM imidazole, 100mM imidazole, and 250mM imidazole solutions were flowed through 4mL of Ni-NTA beads. To remove the imidazole and ensure that the protein was reduced, dialysis was done overnight in 1L of storage buffer (25mM HEPES, 300mM NaCl, 2mM EDTA, 40% (v/v) glycerol, and 1mM ß-mercaptoethanol, ddH2O). The sample was then run through an SDS-PAGE.

Protein Concentration Determination

The concentration of the purified AaCdc14 was determined using the Bio-Rad Bradford Assay with 5x, 10x, 20x, 25x, and 50x dilutions of the protein. The absorbances of the different dilutions were then measured using a spectrophotometer at a wavelength of 595 nm. Using the BSA (bovine serum albumin) standard curve, the absorbances were converted to concentrations and the concentration of the original purified protein was identified.

To confirm the concentration calculated from the Bradford Assay, a nanodrop spectrometry was performed as well. The UV absorbance of the 20x dilution was measured and the concentration of the sample was calculated using the extinction coefficient of AaCdc14 with a His-tag (69790 $M^{-1}cm^{-1}$). The extinction coefficient was found by inputting the amino acid sequence into the ExPASy ProtParam tool. The two concentrations were then compared.

Homology Modeling

The protein sequence for AaCdc14 was inputted into the Molecular Operating Environment (MOE) and modeled after the structure of ScCdc14 (protein code 5XW5) (19). Dissimilar residues from the N-terminus and C-terminus of the AaCdc14 sequence were removed after aligning the two sequences. Utilizing the MOE software, 10 homology models were generated and the model with the lowest RMSD was selected. In some models, the important nucleophilic cysteine residue from the conserved domain was deleted, so the chosen model was checked to make sure the cysteine residue was present.

Ligand Docking & Active Site Interactions

The ligand of the crystal structure of ScCdc14 was tested in MOE against the AaCdc14 homology model. The active site of the two are almost identical, and as such, the ligand of ScCdc14 should be similar to the ligand of AaCdc14 (Fig. 5). The MOE software ran through 20 different possible docking poses and displayed the best 5 according to conformational energy. The best docking pose of those 5, which had the substrate functional groups closest to the optimal substrate position, was saved and a ligand interaction map was generated between the substrate and the protein. The map represents the optimal configuration for binding in the active site of the protein, which includes competitive inhibitors, so this information can be later used in designing a good inhibitor for AaCdc14.

pNPP Activity Assay

Foremost, to determine the linear range for enzyme assays, the enzymes were diluted to 10, 3, 1, 0.3, and 0.1 µM concentrations in the enzyme reaction buffer (25mM HEPES, 1mM EDTA, 150mM NaCl, ddH2O). Then, to set up 6 wells in a 96-well reaction plate, 20 µL of each concentration of enzymes were mixed with 2µL 100mM DTT, 10 µL 800mM pNPP, and 168 µL enzyme reaction buffer in each well, and the last well was a blank with 2µL 100mM DTT, 10 µL 800mM pNPP and 188 µL reaction buffer. The samples were taken to a plate reader to obtain continuous data for 60 minutes, and linear regressions were performed to determine a reaction time and enzyme concentration that guaranteed a linear reaction rate. Then, a standard curve of pNP absorbance against product concentration was created at 405 nm absorbance. The absorbance values of the wells were measured at 405 nm with and without sodium tungstate, the concentrations were determined according to the standard curve, and the specific activity of the enzyme was calculated to be 18.88 pmol pNP/(min*pmol AaCdc14).

pNPP Steady State Assay

To determine the K_M and V_{max} of the enzyme with a pNPP substrate, a pNP standard curve was first generated with concentrations of 1, 10, 30, 50, 60, 80, and 100 µM. An assay was run with varying concentrations of substrate, constant concentration of enzyme, DTT, and a reaction buffer. The assay results were used to create a Michaelis-Menten curve, which was then used with an extension in Google Sheets/Excel, Solver, to find the K_M and V_{max} of the enzyme with a pNPP substrate (Fig. 13).

$$v_0 = \frac{V_{max}[S]}{K_M + [S]}$$

Figure 13 **Michaelis Menten equation**.

Substrate Specificity

The substrate specificity of AaCdc14 was determined using an Enzo Life Sciences Biomol green assay comparing 24 phosphopeptides. 100 µM of phosphopeptides and 0.2uM of AaCdc14 were added to each well along with enzyme reaction buffer and 1mM DTT. 100 µL of Biomol green was used to stop the reaction after 10 minutes and the wells were incubated for 20 minutes. The absorbances of the samples were measured in the

plate reader at 405 nm and converted to concentrations using a Biomol green standard curve.

Inhibitor Assay

An activity assay testing 11 different potential inhibitors of AaCdc14 (Fig. 14) was performed to determine their percent inhibition. A Z factor for the assay was calculated using a positive control (no inhibitor) and negative control (no enzyme) (Fig. 15). The wells testing for inhibitors included 20 µL of inhibitor (each had a final concentration of 100 µM except for H7 and I9, which had a final concentration of 1 mM), 10 µL of 800 mM pNPP, 1 mM DTT, 148 µL of enzyme reaction buffer, and 20 µL of 1.3 µM enzyme. 100 µL of NaOH was used to stop the reaction after 20 minutes. The blank wells contained 20 µL of inhibitor, 10 µL of 800 mM pNPP, 2 µL of 100 mM DTT, and 168 µL of enzyme reaction buffer. The percent inhibition was determined from the absorbance of the inhibitor wells and the blanks detected at 405 nm.

$$Z = 1 - \frac{3(\sigma_p + \sigma_n)}{\mu_p - \mu_n}$$

Figure 15 **Z factor equation**.

Determining IC_{50} of I1

Using the same general setup as the inhibitor assay, except with differing concentrations of inhibitor instead of different inhibitors altogether, an assay was done to determine the IC_{50} value of I1, the best inhibitor for AaCdc14 of the pre-screened ones. Using final inhibitor concentrations of 100 µM, 60 µM, 30 µM, 10 µM, 6 µM, 3 µM, 1 µM, 0.6 µM, 0.3 µM, and 0.1 µM, a four-parameter logistic curve was generated to find the IC_{50} value of the I1 inhibitor (Fig. 16). Volumes from the inhibitor assay were halved to save resources.

$$y = D + \frac{A - D}{1 + 10^{(x - log(c))B}}$$

Figure 16 **Four-parameter logistic curve equation**.

Inhibitor Docking

The 11 different inhibitors (Fig. 14) were docked into the AaCdc14 homology model. The software ran through 20 poses for each inhibitor, and the top 5 based on S score were displayed. An analysis between results from the inhibitor assay, the S score of the different poses, and the interactions of the inhibitor with the protein was used to determine which inhibitor was ideal for *A. a*.

Inhibitor Optimization

An electrostatic map was generated for the docked I1 inhibitor in MOE. Different functional groups were altered and the conformational energy was minimized to view which changes would increase the enzyme affinity for the inhibitor. These changes were based on the polar and nonpolar areas of the enzyme, as predicted by MOE, as well as the original crystal structure of the ScCdc14 (5XW5) substrate (20). Polar functional groups were added to the polar areas to form hydrogen bonds and dipole dipole bonds since those would stabilize the docking. Because MOE is modeling competitive inhibition at the active site, making the inhibitor similar to the original substrate, which the enzyme evolved to fit to, likely would make the inhibitor dock into the enzyme more efficiently.

References

1.. 2020 - hunger map. In: UN World Food Programme. https://www.wfp.org/publications/hunger-map-2020.

2. U.S. and World Population Clock. In: United States Census Bureau. https://www.census.gov/popclock/.

3. Godfray, H. C., Mason-D'Croz, D., & Robinson, S. (2016). Food system consequences of a fungal disease epidemic in a major crop. Philosophical transactions of the Royal Society of London. Series B, Biological sciences, 371(1709), 20150467. https://doi.org/10.1098/rstb.2015.0467

4. Nowicki,M.,Nowakowska,M.,Niezgoda,A. & Kozik,E.(2012).Alternaria Black Spot of Crucifers: Symptoms, Importance of Disease, and Perspectives of Resistance Breeding. Journal of Fruit and Ornamental Plant Research,76(1) 5-19. https://doi.org/10.2478/v10032-012-0001-6

5. Robiglio, A. L., & López, S. E. (1995). Mycotoxin production by Alternaria alternata strains isolated from red delicious apples in Argentina. *International journal of food microbiology*, *24*(3), 413–417. https://doi.org/10.1016/0168-1605(94)00035-5

6. Miles, A.K., Willingham, S.L. & Cooke, A.W. (2005). Field evaluation of a plant activator, captan, chlorothalonil, copper hydroxide, iprodione, mancozeb and strobilurins for the control of citrus brown spot of mandarin. *Australasian Plant Pathology* 34, 63–71. https://doi.org/10.1071/AP04085

7. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 3034285, Azoxystrobin. https://pubchem.ncbi.nlm.nih.gov/compound/3034285.

8. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 9578570, Consist. https://pubchem.ncbi.nlm.nih.gov/compound/9578570.

9. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 86412, Acibenzolar-S-Methyl. https://pubchem.ncbi.nlm.nih.gov/compound/86412.

10. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 37517, Iprodione. https://pubchem.ncbi.nlm.nih.gov/compound/37517.

11. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 15910, Chlorothalonil. https://pubchem.ncbi.nlm.nih.gov/compound/15910.

12. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 8606, Captan. https://pubchem.ncbi.nlm.nih.gov/compound/8606.

13. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 164826, Copper hydroxide. https://pubchem.ncbi.nlm.nih.gov/compound/164826.

14. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 6422843, Pyraclostrobin. https://pubchem.ncbi.nlm.nih.gov/compound/6422843.

15. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 13307026. https://pubchem.ncbi.nlm.nih.gov/compound/13307026.

16. National Center for Biotechnology Information (2022). PubChem Compound Summary for CID 91650, Pyrimethanil. https://pubchem.ncbi.nlm.nih.gov/compound/Pyrimethanil.

17. Li C, Melesse M, Zhang S, et al (2020) FGCDC14 regulates cytokinesis, morphogenesis, and pathogenesis in fusarium graminearum. In: Molecular microbiology. https://pubmed.ncbi.nlm.nih.gov/26256689/.

18. DeMarco A, Milholland K, Pendleton A, et al (2020) Conservation of CDC14 phosphatase specificity in plant fungal pathogens: Implications for antifungal development. In: Scientific reports. https://pubmed.ncbi.nlm.nih.gov/32694511/.

19. Chen N, Uddin B, Voit R, Schiebel E (2015) Human phosphatase CDC14A is recruited to the cell leading ... - pnas.org. In: PNAS. https://www.pnas.org/doi/10.1073/pnas.1515605113.

20. Bank RCSBPD (2017) 5XW5: Crystal structure of budding yeast cdc14p (C283S) bound to a swi6p phosphopeptide. In: RCSB PDB. https://www.rcsb.org/structure/5xw5.

21. Fernández-Ruiz R, Pino M, Hurtado B, et al (2015) Role of sodium tungstate as a potential antiplatelet agent. In: Drug design, development and therapy. https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4454192/. Accessed 19 Jul 2022

22. National Center for Biotechnology Information (2022). PubChem Compound LCSS for CID 2259, Aurintricarboxylic acid. https://pubchem.ncbi.nlm.nih.gov/compound/2259#datasheet=LCSS.

23. Eun H-M (2007) Enzymes and nucleic acids: General principles. In: Enzymology Primer for Recombinant DNA Technology. https://www.sciencedirect.com/science/article/pii/B9780122437403500041.

Supplemental Figures

```
CLUSTAL O(1.2.4) multiple sequence alignment


RcCdc14     -------------------MTPAVKYGQIIEYIPDRLYLASYSSPPSADTLFPYPDPAPS      41
PdCdc14     -----------------M-TTDFNQSGQMIEYIQDRLYLASYDSAPSSRTPFPFHLDAPK      42
AaCdc14     ----------------MS-RASTTSYGQVIEYIQDRLYLASYTSPPDANTPFPYPKQSQS      43
CzCdc14     -------MPHQSRSSVQE-ASAMTSFGQVIEYIQDRLYLASYTHTPDESTPFPYPAKQSR      52
PcCdc14     MADVSTSRPNQPIQPVASGSSAIRTPSLPIAQFGPRLGLTCFSNPIPDPITLNSTHG---      57
UhCdc14     ------------------------MAHPLIRIHKRLWFTYFSGSLPTEDSLNINNGQSA      35
                                     .  :  :  ** :: :         :

RcCdc14     RRSPSKRSARAVDGPSAILSERDRTPPCYFSIDDSLLYNAFHHDFGPLHIGHLYRFAVQF      101
PdCdc14     SPS-KRA----RA--QPA-TPSKRRSPVYFTIDDTLLYNAFHADFGPLHIGHLYRFAVLF      94
AaCdc14     P---SKRSSRAQVGPTA--AGPKAHPPVYFSIDKTLLYNAFHGDFGPLHIGHLYRFAVQL      98
CzCdc14     SPH-KSRSSRAQAGPQSA-AAAPQHPPFYFSIDHTLLYNAFHADFGPLHIGHLYRFAVQL      110
PcCdc14     ------------------FIQSNLQPVYWFTIDSLLIYLSFYQDTGPLNCACLYRFCLHL      99
UhCdc14     A---WKNFTDSEVKKYVDPCETNPKRLHWFNIDHDLVYLSFWFDWGPLNVAMFYRFCLHV      92
                                        :*.**  *:* :*  * ***: . :***.: .

RcCdc14     HDILGAPENKSRPVVFWSKADARSRANAACLLASYMVLIQSWAPHLALAPIAQADPPLMP      161
PdCdc14     HEILGDPANSDRPVVFYSKTDARSRANAACLVACYMVMIQSWPPHLALAPIAQADPPYMP      154
AaCdc14     HEVLGDPANEDRAVVFWSSADSRSRANAACILACYMVLIQSWPPHLALAPIAQMDPPCMP      158
CzCdc14     HEVLGHPDNENRPVVFWSHADSRSRANAACILATYMVLIQNWPPHLALAPIAQMDPPCMP      170
PcCdc14     HELLEDSSMADRRIILYSSDDPDKKANAALLMALYCMIVLRWSVADALHPISHLE--FQP      157
UhCdc14     HHMLNDPDMDAATMVLYTSDDPARKANAALLVTMYAMVIGKCTPADAFYPIAELE--FKP      150
            *.:*         :::::  *   :**** ::: * :::       *: **:. :    *

RcCdc14     FRDAGYSQADYGITVQDVVYGVWRAKEQGFCALSQFDLEEYERYERVDQGDFNWLTPDFL      221
PdCdc14     FRDAGYSQADFILNIQDVVYGVWKAKEQSLCGLRDFNLEEYEKFERVDMGDFNWVTPDFL      214
AaCdc14     FRDAGYSQADYVLNIQDVIYGVWKAKEEGLCGLKDFSLEEYEKFERVDMGDFNWVTPNFL      218
CzCdc14     FRDAGYSQADYSLTIQDVVYGVWKAKEENLVGLKEFSLEEYEKYERVDMGDFNWVSPDFL      230
PcCdc14     FRDAGYSRADFNLSIQNVLFGVKKAIDLRLLKLEEFDLKEYETFEKVEHGDYNWLSPHFI      217
UhCdc14     FRDAGYGRADYSLAIQDVLYGVNRALSEGLLDLSSFNLDEYEKYEQVANGDWNWITPNFI      210
            ******.:**: : :*:*::** :* .  :  * .*.*.*** :*:*  **:**::*.*:

RcCdc14     AFASPQHTPVAPISPTSPLFATLPSTLKAVDAHPTLPSPFKNVLKHFSDRNIGLVVRLNS      281
PdCdc14     AFASPQQQPVAPIPVNTAEYKALPATISEISS-SKLPTPFKNVLVHFHQRNIGLVVRLNS      273
AaCdc14     AFASPQQQPTHEIPTSSPMYATLPSNIAEIQR-SGLPGPFKNVLSHFCARNVGLVVRLNS      277
CzCdc14     AFASPQHLPTHVFQPTEPLYAQLPRTVSAVKRDSNLPTPFKNVLTHFAERGIGLVVRLNS      290
PcCdc14     AFASPVESAN----------GRI-------------GKAFKLIMDQFERVGVKLVIRLNK      254
UhCdc14     AFASPNDRAYVEALRL--GQGRLPPSYS--FRPPEENKLFGKTIRYFKQSGVKLVVRLNN      266
            ***** .              :               *   :  *   .: **:***.

RcCdc14     ELYSPSYFTALGIEHLDMIFDDGTCPPLSTVRKFITLAHEMITVKKRGIAVHCKAGLGRT      341
PdCdc14     ELYCPSYFTAMGIAHVDMIFEDGTCPPLQLVRRFIKMAHEMITIKKKGIAVHCKAGLGRT      333
AaCdc14     ELYSSSYFAKLGIQHLNMIFDDGTCPPLSLVRKFVNLAHEMITVQKRGIAVHCKAGLGRT      337
CzCdc14     ELYSPSYFTALGIKHLDMIFDDGTCPPLNLVKKFISLAHQMINEKRKGIAVHCKAGLGRT      350
PcCdc14     KLYDETRFTKRGIAHREMYFDDGTNPTMEMVREFITISERIIE-EGGVVAVHCKAGLGRT      313
UhCdc14     PLYDRDAFLNAGIDHADMYFDDGSNPTQEILQEFIAKAHRVVS-QGGVVAVHCKAGLGRT      325
             **    *   ** * :* *:**: *  . ::.*:  :..::  :   :***********

RcCdc14     GCLIGAYLIYRYGFTANEIIAYMRFMRPGMVVGPQQHWLHLNQGTFREWWLEEQFEIKMK      401
PdCdc14     GCLIGAYLIYKYGFTANEVIAFMRFMRPGMVVGPQQHWLHLNQGAFREWWFEDSMREKLA      393
AaCdc14     GCLIGAYLIYKHGFTANEIIAYMRFMRPGMVVGPQQHWLHLNQGTFREWWFEDTIKAKYA      397
CzCdc14     GCLIGAYLIYRYGFTANEVIAFMRFMRPGMVVGPQQHWLHLNQGTFREWWFEDTMREKLM      410
PcCdc14     GTLIGAFLIYKYRFTAEEAIGFMRIMRPGTCVGPQQHFLYENQLTWIEWAARDELLADQQ      373
UhCdc14     GVLIGAYLVCKHGFSAGEVIAFMRFMRPGCVVGPQQHFIYQNSLEWVRWGVRDAAMSEAR      385
            * ****:*: :: *:* * *.:**:****  ******::: *.  : .*  .:    .

RcCdc14     EKMANMVPHTPTKLSSKRYLLNSQIATPPNGSR-----RSPL----------GEVD----      442
PdCdc14     QSA-PVTPR----VSTKKRTSNGVVSTPPNNSH---SKRAAL----------GEID----      431
AaCdc14     Q-MMPSTPTK--SPHKQRLASNGQTFTPPNGSSK----RAAL----------GEIE----      436
CzCdc14     ASMQPATPTRLQQSNSRKLASNGQTFTPPNGDRHQTSPRRAL----------GEIT----      456
PcCdc14     AQITPSTTERPITPPPENVVVP-TIALSSATA----------------------------      404
UhCdc14     SVVESERTRL-VD-EVERLKMEANRAVGEGGSKG--VKRRAMEAEESAEEENGEKEAETE      441
                        ..

RcCdc14     ------N------------------EQ-----------RSSIGVQEDCLPAPTPGQPRKT      467
PdCdc14     ------H------------------NEAA-------------AYPDQDLPAPTPGQPRKS      454
AaCdc14     ------S------------------NERS--------NSGHIGVLEDNLPAPTPGQPRKT      464
CzCdc14     ------N------------------NEGAPASTYTDHMKSSVGVADENLPAPTPGQPRKT      492
PcCdc14     -----TP----PPKPLA------------------------S-GSTTMPTSAVPGQPRKT      430
UhCdc14     KGGEERPRQKRKSDPLAIQDAMTPRAQRGSASCIAIHLPATV-GPAGVKPAPCVGQPRKS      500
                                                       :   *****:

RcCdc14     SRAAERHHA  476
PdCdc14     HRKDSRHH-  462
AaCdc14     SKLYGNGRS  473
CzCdc14     NKVYGGRR-  500
PcCdc14     PXHHHHHH-  438
UhCdc14     PSPSRKR--  507
```

Figure 1 **Clustal omega of Cdc14 phosphatase from *A. alternata, P. digitatum, U. hordei, C. zeina, R. commune,* and *P. coronata***. The asterisks indicate residues that are similar between all species. A major conserved domain is AVHCKAGLGRTG, which includes the cysteine and the arginine 6 residues downstream.

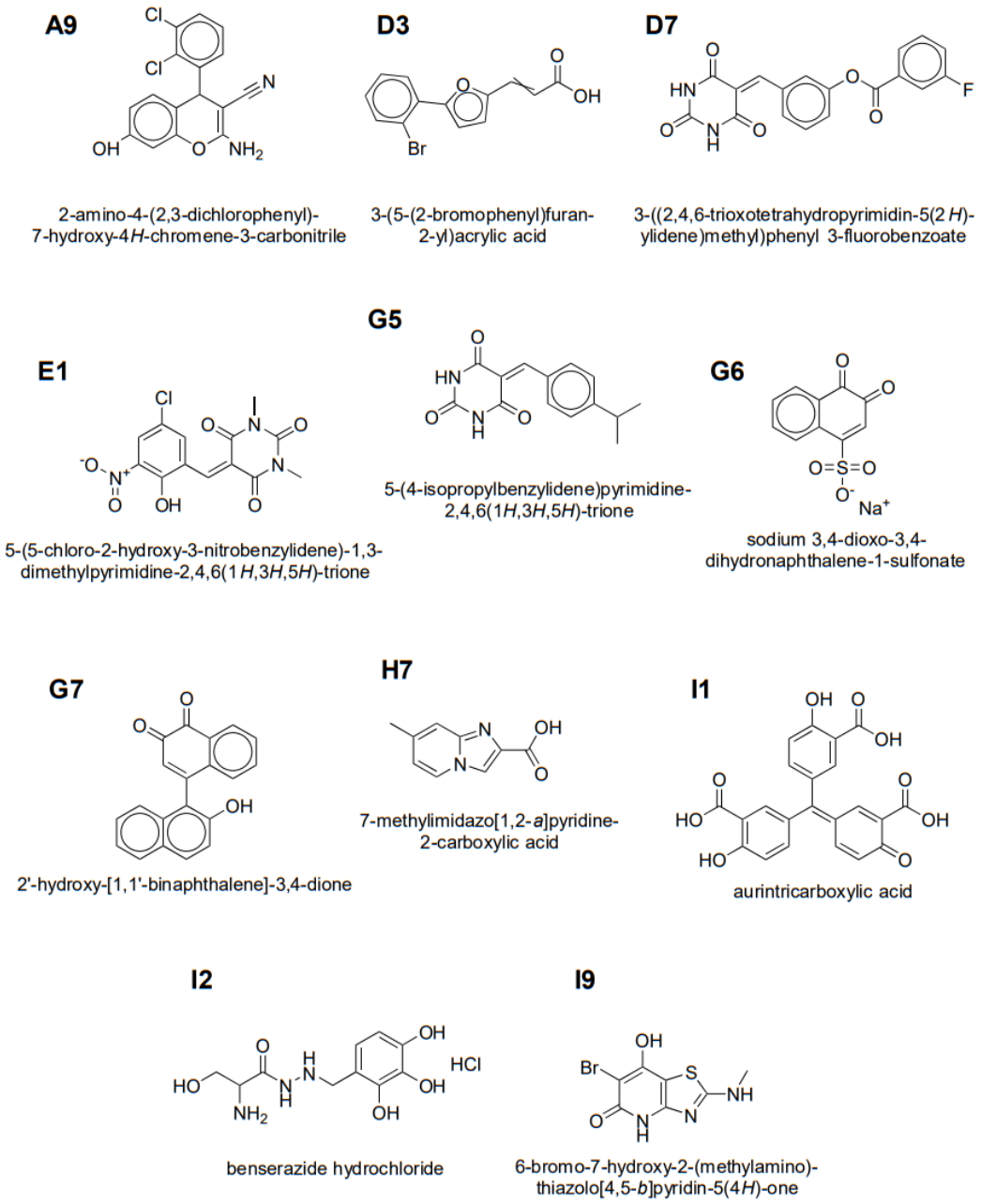

Figure 14 **Inhibitor molecules**. Properties of a good inhibitor include having an atom with high electrophilicity so as to form covalent interactions with the nucleophilic Cys330 and having strong intermolecular interactions to stabilize the docking of the inhibitor. High electrophilicity is indicated, among other factors, by a good carbocation resonance structure, and both polar interactions, like hydrogen bonding and dipole dipole forces, and hydrophobic interactions stabilize the docking.